

Samrat Kumar Dev Sharma

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LinkedIn | ORCID | GitHub | Google Scholar | ResearchGate | Website

Objective

Seeking a PhD position in Statistics or related fields to apply expertise in interpretable machine learning, deep learning, explainable AI (XAI), biostatistics, causal inference, and high-dimensional data analysis.

Education

M.Sc. in Statistics (in progress; expected graduation 2027) | CGPA: 3.60/4.00

Jan 2026 – Present

Jagannath University, Dhaka, Bangladesh

Thesis: Causal Representation Learning for Polygenic Risk Prediction of Alzheimer's Disease Using Multi-Modal Genomic and Neuroimaging Data

Supervised by [Dr. Md. Kamruzzaman](#)

B.Sc. in Statistics | CGPA: 3.65/4.00

Jan 2020 – Aug 2025

Jagannath University, Dhaka, Bangladesh

Thesis: Prevalence and Determinants of Malnutrition Among Women of Reproductive Age in Urban and Rural Bangladesh

Supervised by [Dr. Muhammad Tareq](#)

Technical Skills

Languages: Python, R, C/C++, SQL | **Statistical Software:** SPSS, Stata | **ML/DL Tools:** PyTorch, TensorFlow, Scikit-learn, Keras, HuggingFace, MLflow | **Visualization:** Tableau, Microsoft Power BI | **Bioinformatics:** PLINK, PRSice-2 | **Tools:** Git, GitHub, LaTeX, MATLAB, Microsoft Office

Research Work Experience

Graduate Research Assistant | Jagannath University

Aug 2025 – Present

- **Research Project:** Assessing the Prevalence and Determinants of Hypertension in Bangladesh: Insights from a National Survey Using Machine Learning Techniques.

Field Data Investigator | Jagannath University

Nov 2022 – Feb 2024

- **Research Project:** Psychosocial Problems Among School-Going Adolescents in Dhaka City, Bangladesh.

Reviewer | *Pediatric Research; BMJ Public Health*

2026

Work Experience

Research Assistant | ARI, Dhaka, Bangladesh

Jun 2025 – Present

- Conduct research in public health, machine learning, and bioinformatics; develop deep learning and survival analysis models.

Data Analyst Intern | Techscooper, India (Remote)

Apr 2025 – Jun 2025

- Performed data preprocessing, exploratory analysis, and developed Microsoft Power BI dashboards.

Data Science Mentor

Dec 2024 – Present

- Mentor students in Python, R, Scikit-learn, TensorFlow, and machine learning project development.

Publications

Journal Articles

1. **Sharma, S. K. D.**, Ador, M. Y. H., Rukonuzzaman, M., Chakma, F., Hossen, M., Hossain, J., and Kamruzzaman, M. (2026). Predicting low birth weight in Bangladesh using interpretable machine learning models. *BMC Pregnancy and Childbirth*. <https://doi.org/10.1186/s12884-026-09255-2>
2. Hossain, J., Rouf, A. S. M. R., Tareq, M., Rokunuzzaman, M., **Sharma, S. K. D.** (2026). Assessing the impact of early marriage and socioeconomic determinants on under-five morbidity: A cross-country analysis in South Asia. *BMC Pediatrics*, 26(1), 220. <https://doi.org/10.1186/s12887-026-06596-x>

Preprint

1. **Sharma, S. K. D.** (2024). Oral squamous cell detection using deep learning. *arXiv preprint*, arXiv:2408.08939.

Revised Manuscripts (Resubmitted / Under Review)

1. **Sharma, S. K. D.**, et al. *Survey-Calibrated Machine Learning for Predicting Acute Respiratory Infection in Under-Five Children: Evidence from Bangladesh Demographic and Health Survey 2022*. Revised manuscript resubmitted to *Scientific Reports*.
2. **Sharma, S. K. D.**, et al. *Independent Survival Benefit of Early Stage Diagnosis in Lung Cancer Using Propensity Score Matching and Stratified Cox Proportional Hazards Models*. Under peer review at *Discover Oncology*.

3. **Sharma, S. K. D.**, et al. *Prevalence and Determinants of Diabetes and Prediabetes Among Adults in Bangladesh: A Cumulative Link Mixed Model Approach*. Submitted to *Endocrinology, Diabetes & Metabolism*.
Conference Proceedings

1. **Sharma, S. K. D.**, Nirob, S. M., Riya, R. I., Ador, M. Y. H., Rokunuzzaman, M., and Islam, M. (2026). *Evaluating Breast Cancer Treatment Outcomes Through Machine Learning Survival Analysis and Model Interpretability*. Proceedings of the International Conference on Applied Artificial Intelligence (2AI 2026), Central University of Kashmir, India, in collaboration with the University of South Dakota AI Research (USA). Springer CCIS. *Accepted*.

2. Kabir, M., **Sharma, S. K. D.**, and Kamruzzaman, M. (2025). *Machine Learning Analysis of Adolescent Pregnancy Determinants in Bangladesh*. Proceedings of the 19th National Statistical Conference, Bangladesh Statistical Association (BSA), Bangladesh.

3. **Sharma, S. K. D.**, Ador, M. Y. H., Rokunuzzaman, M., Hossen, M., et al. (2025). *Breast Cancer Detection with Explainable AI: Grad-CAM and Bounding Box-Based Interpretability*. Bangladesh Breast Cancer Conference 2025, Bangladesh.

Honors, Fellowships & Professional Development

NST Fellowship | National Science and Technology Fellowship, Ministry of Science and Technology, Bangladesh 2025–2026

M.Sc. Thesis Fellowship | Jagannath University, Bangladesh 2025–2026

Best Poster Presenter | BIE Environment Workshop 2025 2025

Competitions | Participant, Robi Datathon; [Kaggle ML Competitions](#) 2023–Present

Professional Certifications | Machine Learning Specialization (edX); Neural Networks and Deep Learning (DeepLearning.AI); SQL for Data Science (Coursera); Python and Statistics for Financial Analysis (HKUST); Data Science Math Skills (Duke University); Python – Basic (HackerRank) 2022–2023

Selected Research Projects

Chronic Kidney Disease Stage Prediction | [\[Code\]](#) 2025
Developed machine learning classification models using clinical and laboratory data for early diagnosis and treatment planning.

Breast Cancer Prognosis Using Survival Machine Learning Models | [\[Code\]](#) 2024
Implemented DeepHit, Random Survival Forests, and Cox-based survival models to predict patient prognosis and evaluate model interpretability.

Accident Risk Analysis and Hotspot Identification in the UK | [\[Code\]](#) 2023
Applied principal component analysis (PCA) and unsupervised clustering to identify accident hotspots from UK road traffic data.

Leadership & Professional Service

CEO & Lead Researcher | Advanced Research & Innovation (ARI) Apr 2026 – Present

- Direct research in biostatistics, artificial intelligence, machine learning, and bioinformatics; mentor researchers and coordinate interdisciplinary projects.

President | Statistics Club, Jagannath University Jun 2026 – Present

- Lead student research initiatives by organizing seminars, workshops, hackathons, and networking events in statistics, data science, and AI.

Account Assistant | Birol University Students’ Welfare Association of Dhaka (BUSWAD) May 2020 – Jun 2023

- Assisted in financial record management, budgeting, and administrative activities while supporting student welfare and community service initiatives.

Vice President | Mohana Mangalpur High School Debate Club 2016 – 2017

- Led debate activities, organized inter-school debate competitions, and mentored junior members in public speaking and critical thinking.